

Activity Guide — senior / module-05-dot-addition

IKS Curriculum

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Activity 01 — Base-n Dot Method Drill

Module: Bindu Paddhati (Senior) · **Day:** 5 · **Time:** 15 min within Day 5

Goal

Students apply the dot method to columns in base 8 and base 16, verifying their answers by base-10 conversion.

Materials

- Drill sheet (below)
- Base-conversion reference card

Instructions

Use the dot method. Mark dots. Compute units + carry. Verify by base-10 conversion.

Drill (10 problems)

#	Base	Column
1	8	5, 4, 7, 2
2	8	6, 3, 7, 5, 4
3	8	2, 7, 1, 6, 3
4	16	9, A, 5, 7
5	16	F, 3, 8, A
6	16	7, B, 4, C, 2
7	2	1, 1, 1, 0, 1, 1
8	2	1, 0, 1, 1, 1, 1, 0, 1
9	10	8, 7, 5, 6, 3, 4 (warm-up check)
10	10	4, 9, 7, 2, 8, 1, 6 (warm-up check)

Answer key

#	Units	Carry	Verify (base 10 sum)
1	2 (in 8)	2	$18 = 22_{(8)}$ ✓
2	7 (in 8)	3	$25 = 31_{(8)}$ ✓
3	7 (in 8)	2	$19 = 23_{(8)}$ ✓
4	F (in 16)	1	$31 = 1F_{(16)}$ ✓
5	4 (in 16)	2	$36 = 24_{(16)}$ ✓
6	A (in 16)	2	$42 = 2A_{(16)}$ ✓
7	1 (in 2)	10	$5 = 101_{(2)}$. Units=1, carries=2 (= $10_{(2)}$)
8	0 (in 2)	110	$6 = 110_{(2)}$. Units=0, carries=3 (= $11_{(2)}$)
9	3	3	33 ✓
10	7	3	37 ✓

(Binary problems are tricky — students may need help with the “carry as binary number” interpretation.)

Differentiation

- **Tier up:** Try the binary problems carefully.
- **Tier down:** Stick to base 8.

Clean-up

- Sheets collected for spot-check.

Project Brief

Project — Module 5 (Senior)

Type: 5–6 page research paper · **Marks:** 65 · **Due:** End of Day 10

Choose your track

Math track

Develop a mathematical paper. Options: - Generalise the dot method to base- n and prove correctness via place-value reasoning. - Compare the algorithmic complexity (and constant

factors) of the dot method vs. standard right-to-left addition. - Extend the dot method idea to multiplication (use *Ūrdhva-Tiryagbhyām*) — show how the “spirit” of the technique generalises. - Apply the dot method to non-positional bases (e.g. balanced ternary). Does it still work? Why or why not?

History track

Develop a humanities-style paper. Options: - Argue: “Vedic mathematics” should be relabelled / retained / retained-with-caveat. Defend. - Place Tirthaji’s project in its historical-political moment (Indian independence, post-colonial science, etc.). - Compare Tirthaji’s project with the parallel emergence of Trachtenberg’s system (1930s Germany/Poland). Independent invention or convergent evolution? - Reconstruct what an Atharva-veda *Pariśiṣṭa* containing such sūtras *might* have looked like — and discuss what scholarly evidence would be needed to support the existence claim.

Required elements

- Clear thesis (one paragraph in introduction).
- For math track: at least one correctly-worked algorithm OR formal proof.
- For history track: at least one steel-manned counter-argument with response.
- Citations from at least 3 sources: Tirthaji preface + Dani 1993 + one additional (Plofker, *Vedic Mathematics Batch1.pdf*, Hayashi, etc.).
- Works-cited page in MLA or Chicago format.

Format

- 5–6 pages, double-spaced, 12pt.
- Title page with name, class, track, thesis.

Schedule

- Day 7: choose track.
- Day 8: outline + thesis submitted.
- Day 9: draft (≥2 pages).
- Day 10: paper submitted.

Rubric

See `rubric.md` — different weighting for math vs history tracks.

Self-reflection (1 paragraph)

1. Track and topic chosen + why.
2. Most challenging part.
3. What you’d research deeper with 3 more days.